ATUSHA GHADUSE

atushaghaduse@gmail.com | 9604233589 | Nashik, Maharashtra | Linkedin | GitHub

PROFESSIONAL SUMMARY

Results-driven **R&D Engineer** with expertise in **Data Analytics**, **AI/ML**, and strong skills in **SQL**, **Python**, **Tableau**, and **Power BI** for actionable insight generation. Experienced in EDA, predictive modeling, and dashboard development, with foundational knowledge in **GCP** and **CI/CD**. Passionate about solving business problems through data-driven strategies.

SKILLS

Programming Languages: Python, R, Shell Scripting, YAML, JSON, Linux
Data Analysis & Visualization Tools: Pandas, NumPy, SciPy, Excel, Tableau, Power BI, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)
Data Engineering & Warehousing: ETL Processes, Data Warehousing
Machine Learning & Deep Learning: Regression & Classification Algorithms, Neural Networks, TensorFlow, Keras, PyTorch
Big Data Technologies: Apache Spark (PySpark), Hadoop, HBase, Apache Kafka
Databases: MySQL, MS SQL, MongoDB
Cloud & DevOps: Google Cloud Platform (GCP – Basics), Jenkins (Basics), Docker (Basics), Git (Version Control)

EXPERIENCE

Research & Development Engineer- Data Analyst & ML Engineer

ZentreeLabs Private Limited | Hyderabad, Telangana **July 2024 – February 2025**

- Developed and deployed **interactive Power BI dashboards** by transforming unstructured MongoDB data into actionable insights, reducing report turnaround time by **40%**.
- Established a **seamless integration between MongoDB and Power BI** using ODBC Driver and BI Connector, enabling real-time data visualization for operational and strategic reporting.
- Translated raw data into **structured formats and KPIs**, helping cross-functional teams make informed decisions with embedded dashboards on the application's frontend.
- Conducted data analysis to support **video personalization models**, extracting insights from user interactions and optimizing content generation workflows.
- Collaborated with cross-functional teams to **automate reporting and data workflows**, enhancing data accessibility and decision-making across departments.
- Designed and implemented video processing workflows using OpenCV for high-resolution video creation from image sequences.
- Leveraged OpenCV for advanced computer vision tasks, including image enhancement, frame stitching, and motion effects in video generation.
- Utilized OpenCV's capabilities for image processing and transforming user-uploaded images for seamless integration into video pipelines.
- Automated video creation by integrating computer vision algorithms with machine learning models and advanced Python libraries for personalized video outputs.
- Enhanced user experience by optimizing video processing for performance and quality.
- Collaborated on adding audio tracks to generated videos using the FFmpeg framework in post-production.

AI ML Intern

Cognifront | Nashik, Maharashtra **February 2022 – July 2022**

- Developed a **logistic regression model** to predict match outcomes with **95% accuracy**, enabling actionable insights from historical data for sponsors and analysts.
- Performed comprehensive **data cleaning, preprocessing, and feature engineering**, including null handling, one-hot encoding, and categorical-to-numerical transformations.
- Leveraged **Python (Pandas, NumPy, Scikit-learn)** for exploratory data analysis, model development, and performance optimization using an **80-20 train-test split**.
- Designed a **scalable and efficient classification pipeline**, supporting data-driven decision-making across marketing and operational strategies.

PROJECTS

Bank Analytics: A Holistic Approach to Customer Value & Credit Card Prediction — Sept' 2023 – Feb' 2024

- Implemented a neural network using **TensorFlow** and **Keras**, capturing intricate patterns in the data and enhancing the model's ability to understand complex relationships. This led to a 12% improvement in customer value prediction compared to baseline models.
- Developed and optimized machine learning models using **Random Forest** and **XGBoost** for credit card approval. This optimization resulted in a 20% increase in approval accuracy, providing actionable insights into customers' creditworthiness and reducing potential default risks.

Mental Health and Well-being Surveillance, Assessment & Tracking Solution in Children — July 2022 – June 2023

- Utilized a **MySQL** database to efficiently store and manage diverse data related to students' mental health assessments and observations, reducing data retrieval time by 40%.
- Employed **Hidden Markov Model (HMM)** algorithms to identify sequences of observations and hidden states within the user input, enabling a nuanced understanding of mental health dynamics over time. Achieved a 20% increase in accuracy in predicting mental health trajectories compared to traditional methods.

CERTIFICATIONS

- **Cloud Engineering Track, Data Science & Machine Learning Track** - 30 Days of Google Cloud Program
- **Intellipaat PowerBI Certification Course** - Intellipaat
- **Executive Post Graduate Certification in Data Science & Artificial Intelligence** – IIT Roorkee
- **Google Analytics Beginners & Advanced Certified** - Google
- **Google Digital Unlock – The Fundamentals of Digital Marketing** - Google
- **Microsoft – AI Classroom Series** - Microsoft
- **Core Team member as Google Developer's Students Club (GDSC) as Content Writer and Event Lead of 'ML Study Jams'-** Google – GDSC

EDUCATION

Centre for Development in Advanced Computing (C-DAC)

Post Graduate Diploma in Big Data Analytics **Sept' 2023 – Feb' 2024**

Gokhale Education Society's R. H. Sapat College of Engineering, Nashik

Bachelors of Engineering (B.E.) & Honor's Degree in Computer Engineering & Data Science **August 2019 – July 2023**